

RATE ANALYSIS – "I st class Brickwork in Superstructure"

I st class Brick in Superstructure with 20Cmx10cmx10cm with 1:6 cement sand mortar Unit-1 Cu.m

Given

For 1 Cum, Cement required is =	1440	Kg
1 bag of Cement contained is =	50	Kg
Size of Brick with mortar =	20x10x10	Cm
Standard size of Brick without mortar =	19x9x9	Cm
Cement =	1	
Sand =	6	
Total =	7	

Calculation of Quantity of Brick ,Cement,Sand			SINo	Particulars	Quantity or No	Rate/Cu.M	Amount
Take 10 Cu.M			1	Materials			
Given =	10	Cum		Brick	5000	6	30000
Length of Brick with Mortar =	0.2	M		Sand	3.29	1300	4282.8546
Breadth of Brick with Mortar =	0.1	M		Cement	16	330	5280
Height of Brick with Mortar =	0.1	M		Scaffolding	-	250	250
Numbers of brick required for 10 Cu.m=	5000	No.					
			2	Labors			
Length of Brick without Mortar =	0.19	M		Head Mason	1	466	466
Breadth of Brick without Mortar =	0.09	M		Mason	6	456	2736
Height of Brick without Mortar =	0.09	M		Mazdoor	12	446	5352
Volume of Brick mortar =	2.31	Cu.M		Contenge	-	200	200
Add dry wasteges for volume of mortar =	15.00	Percent		Total =			48566.855
Total Volume of Brick mortar =	2.65	Cu.M		Add 1% of Water Charges =			485.66855
Add dry weight of wet volume =	45.00	Percent		Total =			49052.523
Total Volume of Brick mortar =	3.84	Cu.M		Add 10% of Profit or Loss =			4905.2523
Quantity of cement =	0.55	Cu.M		Grand Total			53957.776
Quantity of cement in Numbers=	0.03	Nos					
Number of Cement Bags required is =	15.81	Nos					
Number of Cement Bags required is =	16	Nos					
Qunatity of Sand required is =	3.29	Cu.M					

